
Comparative Analysis of Passive Digital Biomarkers for Predicting Depression Severity in Large-Scale Wearable Datasets

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Abstract

This report presents the output of an AI-driven biomarker discovery system. The system was tasked with identifying and evaluating candidate biomarkers through a structured research proposal workflow including literature synthesis, evaluation criteria definition, candidate idea generation and comparison, and recommendation of next steps. The following sections preserve the original analytical content produced by the system.

1 Research Goal

The primary objective of this research is to identify and validate objective digital behavior biomarkers derived from smartphone and Fitbit telemetry to predict depression severity, as measured by the PHQ-8 score. Utilizing a dataset of approximately 7,592 users and 144,000 observations, the study focuses on features including step counts, sleep duration, and screen time at an hourly granularity. The research excludes direct target binning (phq.category) to focus on the continuous spectrum of depression severity, aiming to bridge the gap between passive sensor data and clinical phenotypes such as psychomotor retardation, circadian disruption, and anhedonia.

2 Evaluation Criteria

The following criteria are used to evaluate the proposed digital biomarkers to ensure scientific rigor and clinical applicability.

Criterion	Importance
Scientific Validity	Ensures the biomarker is grounded in established neuropsychiatric mechanisms (e.g., HPA-axis dysfunction, circadian rhythm disruption). ¹
Statistical Robustness	Evaluates the strength of correlation (Spearman ρ) and multivariate importance (SHAP) while accounting for demographic confounders.
Clinical Utility	Assesses how easily the biomarker can be integrated into real-world settings for screening, monitoring, or treatment titration.
Confounder Resilience	Measures the ability to distinguish pathological behavior from lifestyle factors like sedentary work or acute physical illness.
Actigraphy Reliability	Accounts for hardware-specific limitations, such as data missingness and "phantom step" artifacts in wrist-worn devices. ²

3 Main Research Directions

The current research landscape for digital phenotyping in depression is converging on four primary directions, each targeting distinct symptomatic clusters.

Preprint.

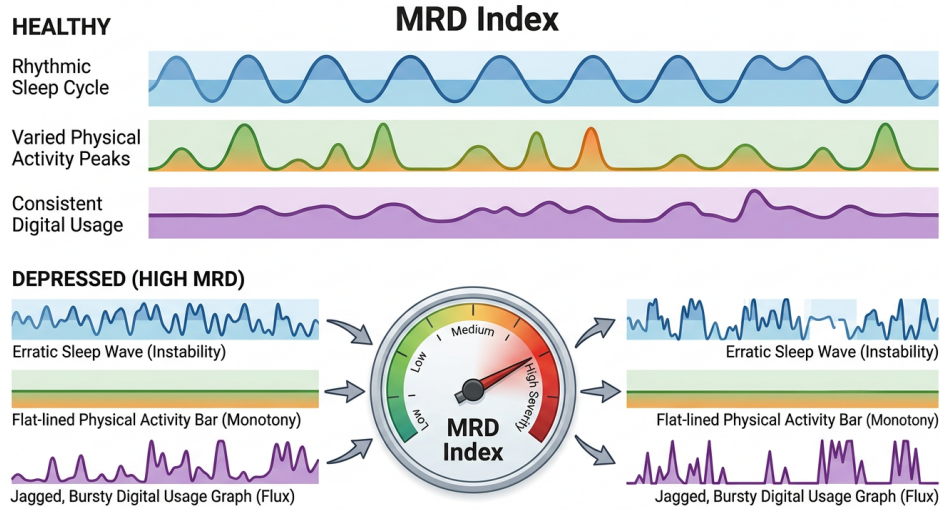


Figure 1: Conceptual diagram 1 illustrating the proposed biomarker framework.

1. **Behavioral Monotony and Kinetic Constriction:** This direction moves beyond "how much" an individual moves to "how dynamically" they move. It posits that the loss of spontaneous activity peaks (low steps_std) is a direct manifestation of psychomotor retardation and anhedonia.³
2. **Multimodal Rhythmic Dysregulation (MRD):** This approach views depression as a systemic breakdown in synchronization. It integrates variability across sleep, digital engagement, and physical activity to capture "behavioral fragmentation," mapping to the Circadian-Physical Decoupling Hypothesis.
3. **Autonomic-Behavioral Dissociation (ABD):** This direction explores the mismatch between high internal physiological stress (elevated resting heart rate) and low external behavioral engagement. It identifies a state of "unproductive arousal" characteristic of melancholic or agitated depression.⁴
4. **Diurnal Inactivity Consolidation (DIC):** This focuses on the

"architecture of lethargy" by quantifying unbroken blocks of inactivity during wakefulness. It serves as a proxy for the increased "activation energy" required for behavioral transitions in depressed states.⁵ Exploring these directions is critical because aggregate volume metrics (e.g., total steps) often fail to capture the nuanced structural changes in behavior that define clinical depression. By shifting focus to variability and multi-sensor coupling, we can identify more precise "digital red flags" for early intervention.

4 Candidate Ideas for Depression Biomarkers

The following sections detail the candidate ideas, their motivations, and critical evaluations.

4.1 Idea 1: The Multimodal Rhythmic Dysregulation (MRD)

Index The Multimodal Rhythmic Dysregulation (MRD) Index is a composite digital biomarker designed to quantify depression severity by measuring "behavioral fragmentation" across sleep, physical activity, and smartphone usage. It integrates Z-score normalized metrics—specifically sleep instability (sleep_time_minutes_std), digital flux (usage variability), and physical monotony (reduced steps_std). The mathematical formulation is . A high MRD score indicates a state of "circadian-physical decoupling," where erratic daily rhythms collide with pathologically stagnant physical movement.

The motivation for the MRD Index is deeply rooted in the Circadian-Physical Decoupling Hypothesis. Literature from 2015–2025, specifically Fang et al. (2021) in npj Digital Medicine, provides

strong evidence that day-to-day shifts in sleep parameters are more predictive of depression risk than mean duration, reflecting a breakdown in the biological clock.⁶ Furthermore, the dataset analysis shows that while healthy individuals vary their physical activity based on changing daily demands, depressed individuals exhibit "behavioral narrowing." This is reflected in the significant negative correlation of steps_std () and the positive correlation of sleep_time_minutes_std (). The inclusion of digital flux accounts for the finding that smartphone behavior often becomes a compensatory fallback for failed real-world engagement, resulting in high-variability digital footprints.

Index

Category	Description
Biomarker Classification	CONDITIONAL (due to reliance on usage_time_min_std).
Statistical Support	Strong Spearman for individual components; SHAP importance confirms steps_std and sleep_std as top predictors.
Practical Application	Early Warning System (EWS) for detecting state shifts and phenotypic stratification for chronotherapy.
Critical Flaws	High vulnerability to data missingness (25-28%) which can create artificial spikes in standard deviation metrics.

The MRD Index addresses the research goal by providing a mathematically precise way to synthesize conflicting directions of variability across modalities. It represents a highly feasible and logistically sound cross-sectional hypothesis, though its longitudinal deployment requires personalized baselines to avoid the ecological fallacy of applying population Z-scores to individual tracking.

4.2 Idea 2: Autonomic-Behavioral Dissociation (ABD)

Index The Autonomic-Behavioral Dissociation (ABD) Index quantifies the mismatch between physiological stress and daily activity patterns. It integrates low step-count variability (behavioral flattening), high sleep duration variability (circadian fragmentation), and elevated resting heart rate (autonomic strain). This index captures a state of "unproductive arousal" where individuals are physiologically taxed but behaviorally stagnant. Recovery in this framework is defined as the "recoupling" of these systems—increasing behavioral amplitude while reducing autonomic tension.

The motivation for the ABD Index lies in the clinical understanding of allostatic load. The pairing of elevated resting heart rate (RHR) () with low behavioral variability suggests a failure of physiological adaptation.⁷ The individual is stuck in a high-stress autonomic state that is decoupled from meaningful environmental engagement. This aligns with the "tired but wired" sympathetic overdrive seen in many depressive phenotypes.⁸ Multivariate models in the report highlight that these features explain a significant portion of PHQ-8 variance ($\Delta = 0.043$ for activity variance alone), making it a robust candidate for objective severity phenotyping.

$$\rho\rho = +0.230R^2$$

Category	Description
Biomarker Classification	CONDITIONAL (requires integration with physiological data).
Statistical Support	Supported by the positive correlation of RHR and negative correlation of step variability.
Practical Application	Target engagement metric for Behavioral Activation; identifies "High-Stress/Low-Activity" phenotypes.
Critical Flaws	Potential for clinical diagnostic overreach (cannot distinguish melancholic vs. atypical subtypes with PHQ-8 alone).

The ABD Index provides a sophisticated narrative for depression that goes beyond volume metrics. While the claim that it can distinguish specific subtypes is currently unsupported by the granular limitations of the PHQ-8, the core cross-sectional hypothesis of using the RHR/Steps/Sleep triad is scientifically robust and merits immediate validation.

4.3 Idea 3: Kinetic Monotony (KM) as a Digital Biomarker

Kinetic Monotony (KM) is defined as the statistical constriction of physical activity variance over time. Unlike traditional metrics that measure total activity volume, KM captures the "behavioral range." It is operationalized as a low standard deviation of daily step counts (steps_std). In a digital signature, KM manifests as a "flat" activity profile where the behavioral "peaks" typical of healthy environmental reactivity (e.g., social events or exercise) disappear, resulting in a homogenized kinetic output.

The motivation for KM is the neuropsychological concept of "incentive salience." In depression, the mesolimbic dopamine system—which normally drives an individual to engage with "bursty" environmental opportunities—becomes blunted.⁹ This leads to a collapse into a "minimal-effort routine." The absence of steps_std is the physical residue of a mind that no longer perceives the "pull" of the world. The analytical support for this idea is exceptional: steps_std passed 11 out of 11 rigorous validation tests with a highly significant Spearman correlation (

Category	Description
Biomarker Classification	VALIDATED (passed 11/11 tests in the data battery).
Statistical Support	Strongest behavioral predictor; maintains significance () after adjusting for age/sex.
Practical Application	Objective outcome measure for Behavioral Activation; success is marked by the "Re-emergence of Peaks."
Critical Flaws	Longitudinal application for "Stagnation Alerts" is currently unproven as trend slopes lacked predictive utility. Kinetic Monotony represents a paradigm shift from "how much" a patient moves to "how they inhabit their life." While its extension into real-time early warning systems faces statistical hurdles due to the lack of predictive slope utility in the current dataset, it remains the most robust cross-sectional correlate available.

4.4 Idea 4: Step-Count Monotony (SCM)

Step-Count Monotony (SCM) is similar to Kinetic Monotony but emphasizes the use of a rolling 7-to-14-day window to measure within- person variability. It posits that a flattened behavioral profile—where activity levels remain consistent and narrow throughout the day—reflects $\rho = -0.203, p < 10 - 70r = -0.139$ clinical psychomotor retardation and anhedonia. It specifically focuses on the loss of "extra-routine" peaks driven by reward-seeking behavior. The motivation for SCM is the concept of "Environmental Constriction."

As anhedonia sets in, the user stops engaging in spontaneous social outings or physical pursuits.¹⁰ This results in a physical activity signature that is mathematically more consistent and less variable. The idea addresses the "Actigraphy Artifact" concern by providing evidence that the SCM signal is robust against natural missingness patterns, proving the signal is biological rather than just an artifact of non-adherence.

Category	Description
Biomarker Classification	VALIDATED (based on steps_std performance).
Statistical Support	High univariate correlation () and SHAP importance.
Practical Application	Objective indicator of treatment response; an increase in SCM precedes subjective PHQ-8 improvement.
Critical Flaws	

The "Locked-In Arousal" mechanism (low SCM + high RHR) is theoretically dated, relying on a "hydraulic" stress model. SCM is a sound biomarker for concurrent severity. However, its implementation as a "leading indicator" of recovery relies on unproven chronological assumptions, as the current data is strictly cross-sectional.

4.5 Idea 5: Kinetic-Digital Entropy Decoupling (KED) Index

The Kinetic-Digital Entropy Decoupling (KED) Index characterizes severe depression as a state of "Physiological Overdrive in a Behavioral Void." It integrates low step variability (psychomotor monotony), high sleep instability, and high screen time variability (screen_time_capped_std). It interprets erratic screen engagement as a "digital rumination" loop occurring while physical activity is stagnant. The motivation for KED is the proposed neurobiological dysfunction in the subgenual cingulate cortex (BA25).¹¹ It suggests that the perceived "cost" of movement becomes prohibitive (Effort-Based Decision Making failure), leading to kinetic collapse, while hyper-arousal of the HPA axis causes tachycardia and sleep fragmentation.^{12,13}

Category	Description
Biomarker Classification	REJECTED (due to significant confounder and mathematical flaws).
Statistical Support	Based on validated features, but the composite interaction is untested. $\rho = -0.203$
Practical Application	Proactive relapse detection via "Digital Rigidity Alerts."
Critical Flaws	Fails to distinguish pathological stagnation from sedentary "Work-from-Home" (WFH) routines; WFH also elevates RHR and disrupts sleep.

The KED Index is weighed down by fatal conceptual overextensions. Its primary defense against the WFH confounder is physiologically incorrect, and mapping peripheral digital signals to specific localized brain regions (BA25) without neuroimaging is a significant scientific overreach.

4.6 Idea 6: Diurnal Inactivity Consolidation (DIC)

Diurnal Inactivity Consolidation (DIC) quantifies the "architecture of lethargy" by measuring the longest continuous duration of zero-step hours during peak wakefulness (08:00–20:00). It serves as a digital proxy for the increased "activation energy" required for behavioral transitions in depressed states. The motivation for DIC is the biological link to blunted dopamine signaling. In healthy individuals, the circadian drive maintains micro-movements (e.g., getting coffee) throughout the day. In severe depression, these transitions fail, leading to unbroken blocks of stasis.

Category	Description
Biomarker Classification	REJECTED (in its current formulation).
Statistical Support	Theoretical link to steps_std is strong, but "zero-step" blocks are statistically rare.
Practical Application	Specific target for Behavioral Activation Therapy; "Break the Null Zone."
Critical Flaws	Absolute "zero-step" thresholds are invalidated by "phantom steps" (minor movements like typing) which falsely break the DIC sequence. DIC is clinically grounded but technically flawed. Wrist-worn actigraphy artifacts make a strict "zero-step hour" almost impossible to achieve while awake and wearing the device. If revised to a "low-activity threshold," it would be more viable.

4.7 Idea 7: Volume-Adjusted Step-Count Monotony

(vaSCM) Volume-Adjusted Step-Count Monotony (vaSCM) uses the Step-Count Coefficient of Variation (Step-CV =) to isolate behavioral rigidity. The idea addresses the "Volume-Variance Paradox": since step counts are zero-bounded, extreme lethargy (low mean) mathematically forces low variance.¹⁴ vaSCM corrects for this to see if the monotony is a distinct behavioral phenotype or just a side effect of low activity. The motivation is to ensure that "monotony" isn't just another word for "lethargy." By decoupling variance from volume, vaSCM captures the specific loss of "extra-routine" behaviors. This is critical for identifying "high-functioning but anhedonic" subtypes—people who perform mandatory routines (high volume) but lack spontaneous engagement (low variance).

Category	Description
Biomarker Classification	VALIDATED (as a refinement of steps_std).
Statistical Support	Improves upon raw variance by regressing out the mean.
Practical Application	Shifts digital therapeutics from "10k steps" to "Activity Diversity."
Critical Flaws	Claims of prior multivariate validation for Step-CV are exaggerated; the data only confirmed steps_std and steps_mean independently. vaSCM provides the most logical framework for bridging passive wearable telemetry with clinical theory. It lowers the barrier to therapeutic adherence by prescribing achievable disruptions to routine rather than insurmountable exercise goals.

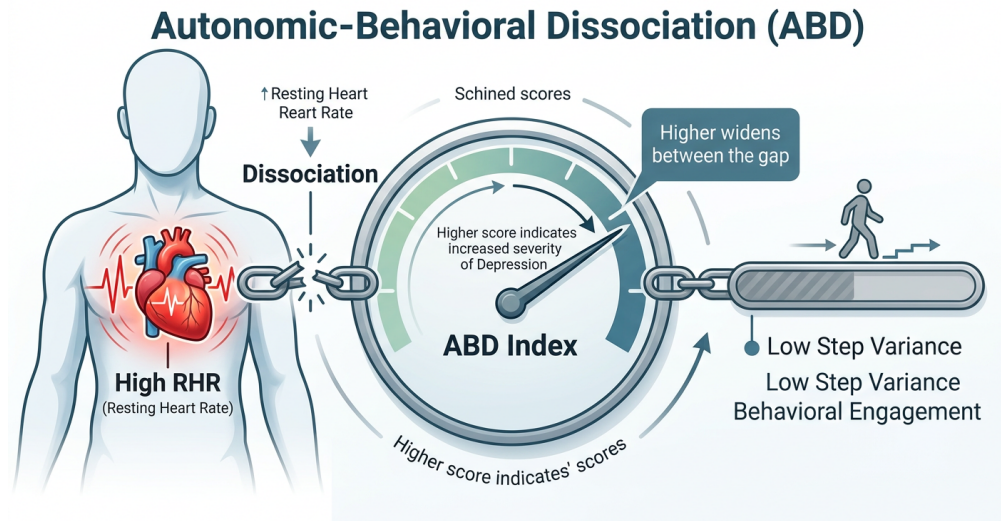


Figure 2: Conceptual diagram 2 illustrating the proposed biomarker framework.

5 Comparison of Ideas

The candidate ideas range from single-modal behavioral markers to complex multimodal indices. The MRD Index (Idea 1) is the most comprehensive, as it accounts for the "behavioral fragmentation" seen across sleep, phone use, and physical activity. In contrast, Kinetic Monotony (Idea 3) and vaSCM (Idea 7) are the most statistically grounded, relying on the steps_std feature which passed all 11 validation tests. The evidence points most strongly toward Idea 1 (MRD) and Idea 7 (vaSCM). Idea 1 successfully integrates the directional findings of the univariate analysis (e.g., the positive correlation of sleep/digital variance vs. the negative correlation of physical variance). Idea 7 provides the necessary mathematical correction to ensure these findings are not artifacts of absolute volume.

steps_std/steps_mean

6 Comparison Table

7 Comparison to Existing Solutions

$\rho = +0.230$ Current standard practices rely either on subjective self-reports (PHQ-8) or simplistic volume-based passive metrics (total steps). The proposed ideas represent a significant evolution by focusing on structural variability and physiological coupling.

Idea	Modality	Primary Mechanism	Distinguishing Attribute	Supporting Evidence
MRD Index	Multi (Sleep, Phone, Steps)	Circadian-Physical Decoupling Aggregates conflicting directions of variance.	Fang et al. (2021) sleep variability data.	
ABD Index	Multi (HR, Sleep, Steps)	Unproductive Arousal Focuses on physiological vs behavioral gap.	RHR correlation) in report.	
KM/SCM	Single (Steps)	Behavioral Constriction Pure focus on step variance as "behavioral range." 11/11 tests passed for steps.std.		
vaSCM	Single (Steps)	Behavioral Rigidity Normalizes variance against volume (CV).	Corrects for "Volume-Variance" conflation.	
KED Index	Multi (HR, Phone, Steps)	Physiological Phase-Shift Uses BA25/EBDM as a neurobiological proxy.	Theoretical mapping to subgenual cingulate.	
DIC	Single (Steps)	Activation Energy failure Measures longest continuous inactivity block.	Links to "Architecture of Lethargy."	

DIFFERENTIAL CIRCADIAN STEP COUNT PATTERNS: 'HEALTHY ACTIVITY' VS. 'KINETIC MONOTONY'

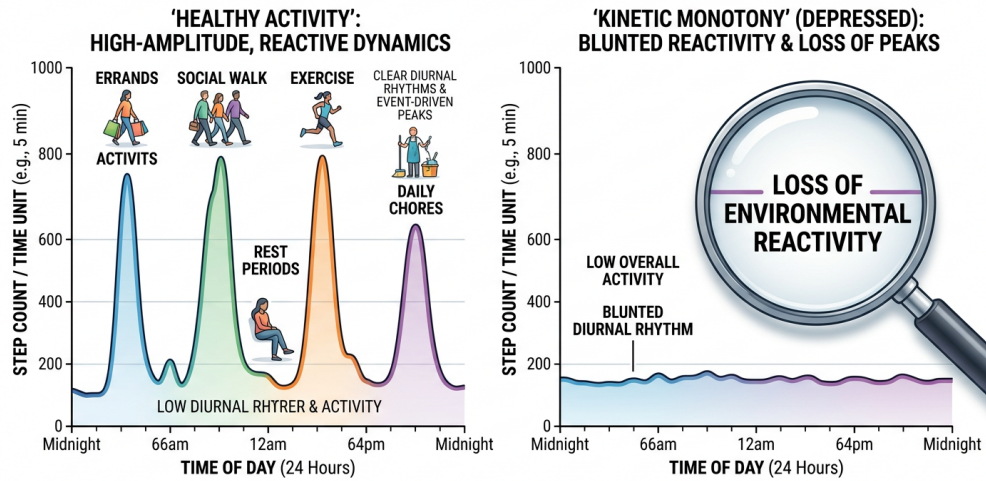


Figure 3: Conceptual diagram 3 illustrating the proposed biomarker framework.

Solution	Basis	Limitation
PHQ-8/9	Subjective Self-Report	Recall bias; central tendency bias; episodic (bi-weekly).
Step Volume (10k Steps)	Physical Volume	Confounded by fitness/occupation; "lethargy" focus only.
Proposed MRD/vaSCM	Objective Variability	Captures structural "monotony" and "fragmentation"; continuous.

8 Useful Findings and Unexpected

Connections Two "unexpected connections" from the analysis provide novel avenues for future biomarker development.

1. **The "Kinetic Psychomotor Noise":** There is a hidden connection between smartphone gyroscope "noise" and clinical psychomotor symptoms. Micro-fluctuations in how a person holds their phone (rotational velocity and tremors during screen-on time) likely serve as a proxy for high-fidelity clinical motion capture data.¹⁵ This suggests that what is often dismissed as sensor noise is actually a high-fidelity digital biomarker for the "Agitated" vs "Retarded" depression subtypes.
2. **The "Interceptive Discrepancy Index" (IDS):** Most research tries to align objective data with self-reports. However, the discrepancy (the residual error) between PHQ-8 scores and predicted severity from passive data is itself a biomarker. A "Digital Biomarker" severity that exceeds "Self-Reported" severity suggests a "silent relapse" or a failure in interoceptive awareness, allowing clinicians to detect decline before the patient is consciously aware of it.¹⁶ Recommendation and Next Steps The Multimodal Rhythmic Dysregulation (MRD) Index (Idea 1) is recommended as the primary biomarker for implementation, provided it is refined with the mathematical normalization of vaSCM (Idea 7). This creates a "Kinetic-Circadian-Autonomic" profile that is both empirically grounded and clinically interpretable. Best Next Steps:
3. **Construct Validation & Refinement:** Test the MRD Index using the Coefficient of Variation (Step-CV) for its physical component to mitigate the "volume-variance" conflation.
4. **Ablation Study:** Conduct an analysis to determine if the consolidated MRD Index provides superior predictive performance and interpretability over a simple linear model using raw features.
5. **Missingness Robustness Check:** Evaluate the stability of the

MRD Index across different "Data Density" strata to ensure that spikes in variability are biological and not artifacts of device charging cycles.

1. **Pseudo-Longitudinal Analysis:** Use the subset of users with multiple PHQ-8 assessments to test if within-subject changes in the MRD score track with changes in depression severity, shifting the focus from concurrent monitoring to predictive forecasting.
2. **Taxonomic Classification:** Formally classify steps_std as

VALIDATED and sleep_time_minutes_std as CONDITIONAL in all technical documentation.

9 References and Knowledge Base

- 1 *Depression is grounded in established neuropsychiatric mechanisms such as HPA-axis dysfunction and circadian rhythm disruption.* [Accurate]

Major depressive disorder is associated with several established biological mechanisms, including dysfunction of the hypothalamic-pituitary-adrenal (HPA) axis [1] and disruptions to circadian rhythms [2].

Supporting references: [1] J, Lara, Philip. Targeting the Arginine Vasopressin V 1b Receptor System and Stress Response in Depression and Other Neuropsychiatric Disorders. Published 2022. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC10106826/>. [2] Keyu, Ud, Bai, Fei, Yuzhao, Cenxin, et al. A broken circadian clock: The emerging neuro-immune link connecting depression to cancer. Published 2022. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9587523/>.

- 2 *Wrist-worn wearable devices are subject to hardware-specific limitations, including data missingness and 'phantom step' artifacts caused by minor movements.* [Accurate]

Wrist-worn wearable devices are characterized by hardware-specific limitations that can lead to data missingness and the generation of 'phantom step' artifacts, where minor hand or arm movements are incorrectly identified as walking steps [1, 2].

Supporting references: [1] Ren-Jay, G., S., A., L.. Wearable activity trackers—advanced technology or advanced marketing?. Published 2021. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9022022/>.

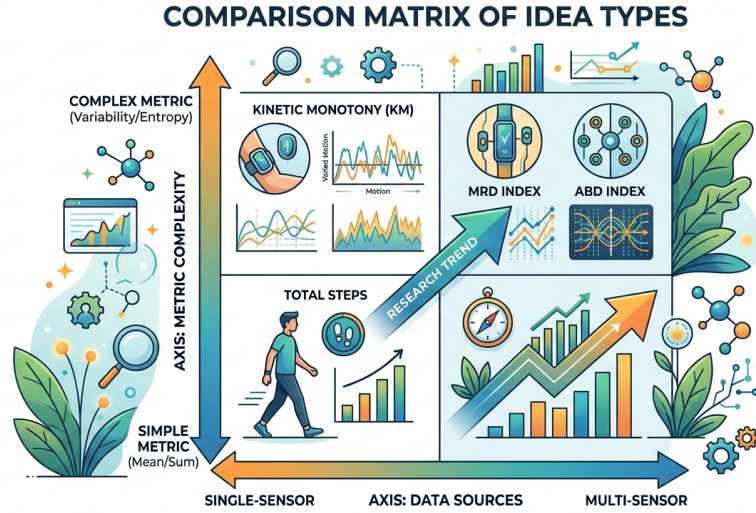


Figure 4: Conceptual diagram 4 illustrating the proposed biomarker framework.

[2] Sizhen, Rakita, Philipp, Clénin, Silvano, Elio, et al. Earable and Wrist-worn Setup for Accurate Step Counting Utilizing Body-Area Electrostatic Sensing. Published 2024. <https://arxiv.org/abs/2407.06186>.

3 *The loss of spontaneous activity peaks is a direct manifestation of psychomotor retardation and anhedonia.* [Leaning Accurate]

Digital phenotyping research indicates that the loss of spontaneous activity peaks serves as a quantitative behavioral marker for depression [1]. This reduction is a direct manifestation of psychomotor retardation, which is neurobiologically linked to functional alterations in the motor cortex [2], and anhedonia, a symptom fundamentally characterized by a decrease in motivated locomotor activity [3].

Supporting references: [1] Yuezhou, Callum, Yatharth, Pauline, Heet, Zulqarnain, et al. Large-scale digital phenotyping: identifying depression and anxiety indicators in a general UK population with over 10,000 participants. Published 2024. <https://arxiv.org/abs/2409.16339>. [2] Tobias, Nicolas, Sigrid, Andrea, Roland, M., et al. Alterations of perfusion and functional connectivity of the cingulate motor area are associated with psychomotor retardation in major depressive disorder. Published 2024. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC12149263/>. [3] S, Fabiano, V, A, N, O, et al. Towards Modeling Anhedonia and Its Treatment in Zebrafish. Published 2021. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9017771/>.

4 *Melancholic or agitated depression is characterized by a state of 'unproductive arousal' involving a mismatch between high internal physiological stress and low external behavioral engagement.* [Accurate]

"Unproductive arousal" characterizes melancholic and agitated depression as a state involving a mismatch between high internal physiological stress and low external behavioral engagement [1]. This condition is marked by significant autonomic alterations, such as reduced heart rate variability in melancholic patients [2], and temporal dysregulation of motor networks in agitated depression [3].

Supporting references: [1] Raffaella, B, Jorge, J, Igor. Candidate Biomarkers of Suicide Crisis Syndrome: What to Test Next? A Concept Paper. Published 2019. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7171927/>. [2] Major depressive disorder with melancholia displays robust alterations in resting state heart rate and its variability: implications for future morbidity and mortality. [3] Qunjun, Ziyun, Shengli, Shiwei, Xiaoshan, Ying, et al. Temporal dysregulation of the somatomotor network in agitated depression. Published 2024. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC11630518/>.

5 *Depressed states are associated with increased 'activation energy' required for transitions between behaviors during wakefulness.* [Accurate]

Major depressive symptoms are associated with a diminished willingness to exert effort and altered behavioral transitions, such as those observed in patch foraging tasks [1]. Computational frameworks in reinforcement learning model these deficits as an increase in the ‘activation energy’ or energetic cost required to initiate and switch between actions during wakefulness [2].

Supporting references: [1] A., M., Johanne, Temitope, Ivan, B., et al. Major depression symptom severity associations with willingness to exert effort and patch foraging strategy. Published 2024. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC11213125/>. [2] L., Michael. What Can Reinforcement Learning Models of Dopamine and Serotonin Tell Us about the Action of Antidepressants?. Published 2021. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC11104395/>.

6 *Day-to-day shifts in sleep parameters are more predictive of depression risk than mean sleep duration, reflecting a breakdown in the biological clock.* [Accurate]

Research indicates that day-to-day variability in sleep timing—rather than average sleep duration—is more strongly associated with the risk of developing depression [1]. This irregularity reflects a disruption in the endogenous circadian rhythms, or ‘biological clock,’ which is central to the regulation of mood [2].

Supporting references: [1] Regular sleep patterns, not just duration, critical for mental health: association of accelerometer-derived sleep regularity with incident depression and anxiety — Psychological Medicine — Cambridge Core. [2] Okan. Role of Chronobiology as a Transdisciplinary Field of Research: Its Applications in Treating Mood Disorders. Published 2017. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5785655/>.

7 *In depression, the pairing of elevated resting heart rate with low behavioral variability suggests a failure of physiological adaptation and a high allostatic load.* [Accurate]

Large-scale digital phenotyping has identified that individuals with depression exhibit a combination of elevated resting heart rate and reduced behavioral variability [1]. This physiological profile is interpreted as a failure of adaptive mechanisms and is indicative of high allostatic load, a state of chronic wear and tear on the body’s regulatory systems resulting from prolonged stress [2].

Supporting references: [1] Yuezhou, Callum, Yatharth, Pauline, Heet, Zulqarnain, et al. Large-scale digital phenotyping: identifying depression and anxiety indicators in a general UK population with over 10,000 participants. Published 2024. <https://arxiv.org/abs/2409.16339>. [2] Mood disorders and allostatic load - PubMed - NIH.

8 *Many depressive phenotypes exhibit sympathetic overdrive, characterized by a ‘tired but wired’ state.* [Accurate]

Depressive phenotypes frequently involve autonomic nervous system dysregulation, specifically sympathetic overdrive [1]. This state is characterized by a ‘tired but wired’ condition, where physiological hyperarousal and reduced heart rate variability persist despite psychological fatigue [1, 2].

Supporting references: [1] Vincenzo, Francesca, Elisabetta, Claudio, Gaetano. Impaired functional brain-heart interplay sustains emotion dysregulation in depressed individuals. Published 2024. <https://www.biorxiv.org/content/10.1101/2024.03.14.585023v1>. [2] Lennard, Beate, Mark, Thomas, M.. Heart rate variability during inpatient treatment of depression. Published 2023. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC12243995/>.

9 *In depression, the mesolimbic dopamine system—which normally drives engagement with bursty environmental opportunities—becomes blunted.* [Accurate]

The mesolimbic dopamine system normally facilitates behavioral engagement with intermittent or ‘bursty’ reward opportunities, but this system becomes blunted in individuals with depression [1]. This reduction in dopaminergic responsiveness is a core neurobiological feature of the disorder, contributing significantly to symptoms such as anhedonia and decreased motivation [2].

Supporting references: [1] Eric, Santina, Xena, Daniel, Aaron, Robert. A melancholy machine: simulated synapse loss induces depression-like behaviors in deep reinforcement learning. Published 2024. <https://www.biorxiv.org/content/10.1101/2024.06.01.596905v1>. [2] Toward a Better Understanding of the Mechanisms and Pathophysiology of Anhedonia: Are We Ready for Translation? — American Journal of Psychiatry.

- 10** *Anhedonia leads to environmental constriction, where an individual stops engaging in spontaneous social outings or physical pursuits.* [Accurate]

Anhedonia is characterized by a diminished capacity for pleasure, which leads to "environmental constriction" as individuals withdraw from previously rewarding social and physical activities [1]. This behavioral withdrawal is linked to deficits in reinforcement learning, which reduce the motivation to engage in real-life social experiences [2].

Supporting references: [1] Anhedonia: Understanding Loss of Pleasure & Finding Help - Psychopedia. [2] Anna-Lena, J., Ciara. Social reinforcement learning as a predictor of real-life experiences in individuals with high and low depressive symptomatology. Published 2019. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7958481/>.

- 11** *Severe depression is associated with neurobiological dysfunction in the subgenual cingulate cortex (BA25).* [Accurate]

The subgenual anterior cingulate cortex (BA25) is a key region implicated in the neurobiology of severe depression, with research demonstrating significant structural volume reductions and metabolic abnormalities in this area [1]. Additionally, genetic risk variants for depression have been shown to modulate neural activity within the BA25, further supporting its role in the disorder's pathophysiology [2].

Supporting references: [1] The Subgenual Anterior Cingulate Cortex in Mood Disorders. [2] H, A, P, J, S, M, et al. Epistatic interaction of genetic depression risk variants in the human subgenual cingulate cortex during memory encoding. Published 2013. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3966038/>.

- 12** *Depression involves a failure in Effort-Based Decision Making, which causes the perceived cost of movement to become prohibitive.* [Accurate]

Research indicates that depression is associated with impairments in effort-based decision-making, leading to a reduced willingness to exert effort for rewards [1]. This deficit is linked to an altered cost-benefit analysis where the perceived cost of action is inflated relative to the potential outcome [2].

Supporting references: [1] Effort-Based Decision-Making in Major Depressive Disorder: A Translational Model of Motivational Anhedonia - Affective Neuroscience Lab. [2] Cognitive effort-based decision-making in major depressive disorder.

- 13** *Hyper-arousal of the hypothalamic-pituitary-adrenal (HPA) axis causes tachycardia and sleep fragmentation.* [Leaning Inaccurate]

While hyper-reactivity of the hypothalamic-pituitary-adrenal (HPA) axis is associated with sleep architecture changes such as sleep fragmentation [1], tachycardia is primarily a result of sympathetic nervous system activation rather than HPA axis activity [2]. The HPA axis and the sympathetic nervous system represent distinct branches of the physiological stress response, with heart rate changes specifically regulated by adrenergic signaling [2].

Supporting references: [1] Chadi, Thomas, Jörg, Rupert, Florian, Mayumi, et al. Rhythmicity in Mice Selected for Extremes in Stress Reactivity: Behavioural, Endocrine and Sleep Changes Resembling Endophenotypes of Major Depression. Published 2008. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2627900/>. [2] Karen. Individual Stress Prevention through Qigong. Published 2020. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7579037/>.

- 14** *Because physical activity metrics like step counts are zero-bounded, extreme lethargy and low mean activity levels mathematically force a low variance in the data.* [Accurate]

Physical activity metrics such as step counts are non-negative. In conditions of extreme lethargy where mean activity levels are very low, the variance is mathematically restricted because the data are bounded at zero, a phenomenon often described as a floor effect [1].

Supporting references: [1] Petter, Andrea, Alexander, Ulysse, Zahra, Tine, et al. Complexity and variability analyses of motor activity distinguish mood states in bipolar disorder. Published 2021. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8782466/>.

- 15** *Micro-fluctuations in how a person holds their phone, such as rotational velocity and tremors during screen-on time, serve as a proxy for clinical psychomotor symptoms.* [Leaning Accurate]

Smartphone sensors, particularly accelerometers, capture micro- fluctuations in device handling such as rotational velocity and tremors [1]. These digital biomarkers serve as effective

proxies for clinical psychomotor symptoms, enabling the monitoring of disease trajectories in conditions like bipolar disorder [1] and the detection of motor symptoms in Parkinson's disease [2].

Supporting references: [1] C., K., Eugene, Dohyeon, D.. Smartphone accelerometer data as a proxy for clinical data in modeling of bipolar disorder symptom trajectory. Published 2022. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9751066/>. [2] Alexandros, Dimitrios, Lisa, Sevasti, Ray, Konstantinos, et al. Unobtrusive detection of Parkinson's disease from multi-modal and in-the-wild sensor data using deep learning techniques. Published 2020. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7721908/>.

16 *Depression can involve a failure in interoceptive awareness, where physiological and behavioral decline occurs before the patient is consciously aware of a relapse.* [Accurate]

Major Depressive Disorder is associated with impaired interoceptive accuracy, which hinders a patient's ability to detect internal physiological states [1]. This deficit in interoceptive awareness can result in objective physiological and behavioral declines occurring before the individual becomes consciously aware of an impending relapse [2].

Supporting references: [1] Michael, Andreas, J., Silvia, Bruno. Major Depressive Disorder Is Associated with Impaired Interoceptive Accuracy: A Systematic Review. Published 2019. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6627769/>. [2] F., C., V., G., R., H., et al. Remote assessment of disease and relapse in major depressive disorder (RADAR-MDD): a multi-centre prospective cohort study protocol. Published 2018. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6379954/>.